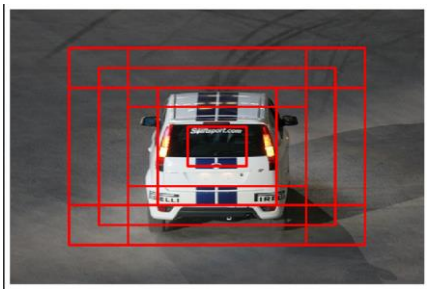
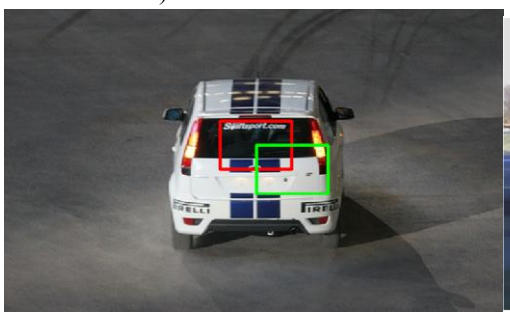
1 *Object Detection*

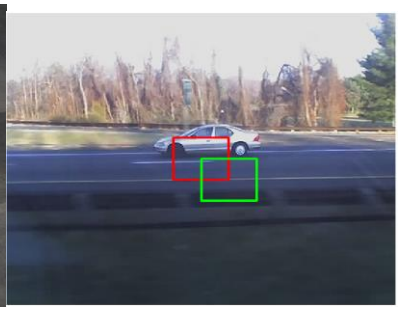
- Attach one image showing the shape of the 9 anchors centered at the center position (Anchor Generator)



The nine anchor boxes are correctly centered and only differ in scale and aspect ratio, confirming proper anchor generation.

- Attach two images showing the result of the (0.5,0.5,0,0) and (0,0,1,1) anchor transformations (Proposal Generator)





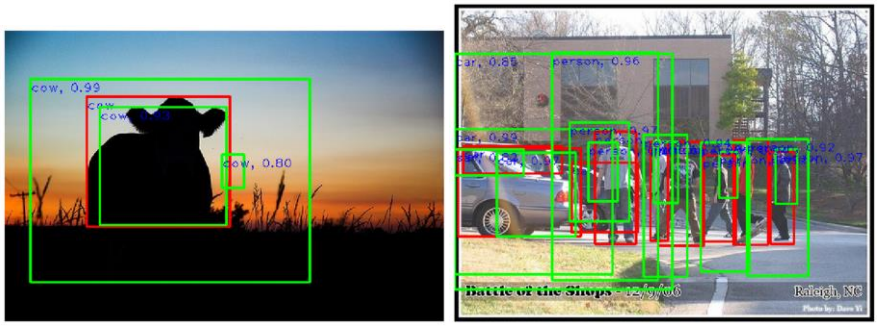
The two visualizations show how the proposal generator shifts or rescales each anchor: (0.5, 0.5, 0, 0) moves the anchor center by half a cell without changing its size, while (0, 0, 1, 1) keeps the center fixed but expands the box by a factor of e.

- For the test using the regular YOLO loss and 50 epochs
    - Attach the detection examples for the overfitted model





    - Attach the detection examples for the model trained in the complete dataset



- By visual inspection of the results (complete training model) analyze the cause/s of the limited performance of the model. In your analysis, consider the existence of FP and FN, IoU between proposals and GT boxes, and correct classification of boxes (this may include objectness and multiple-class classification). Add visual examples to illustrate your discussion. To that end, you can increase the number of inference images by changing the value of *num_sample*, and re-running the definition of *small_dataset* (post-training). Given the conclusions of this analysis, justify which of the following approaches you consider the main way to improve the performance of the detector.
    - Improve objectness classification accuracy
    - Improve multi-class classification accuracy,
    - Improve bbox regression

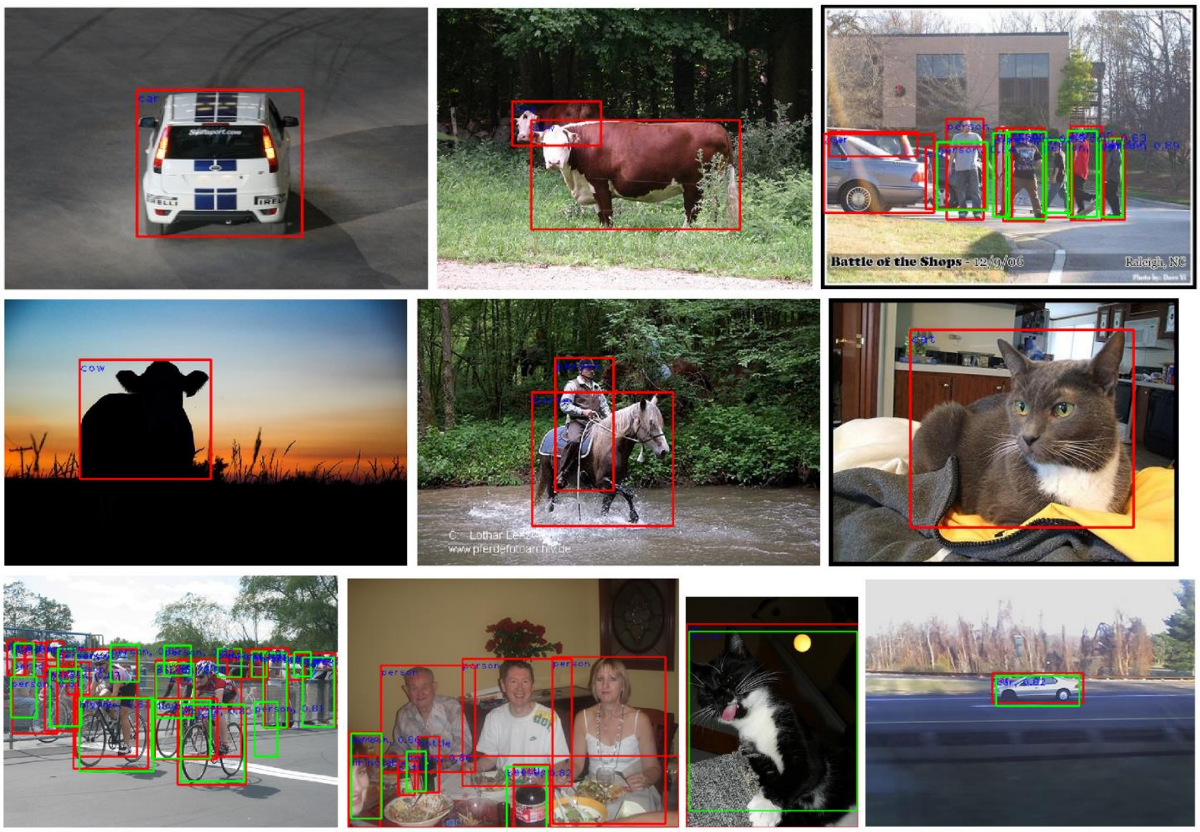
Objectness classification determines whether an anchor truly contains an object, and errors here lead directly to false positives when background regions are mistaken for objects or to false negatives when real objects are ignored; improving it requires better feature representation and more balanced training. Multi-class classification decides which category the detected object belongs to, so mistakes at this stage cause correct boxes to be labeled with the wrong class; this can be improved through stronger supervision, more expressive classifiers, or better training data. Bounding-box regression controls how accurately the box aligns with the object's shape and position, meaning poor regression produces misaligned, oversized, or undersized boxes; enhancing this component involves refining anchor design, stabilizing regression targets, or increasing model capacity to improve localization precision.

For the visual inspection of the detections, the previous images were analyzed. The images reinforce why the overfitted model performs well while the fully trained model struggles. In the cow image, the silhouette is entirely black and lacks color cues, yet the model classifies it correctly, suggesting that color variation does not strongly affect class prediction; however, the full-dataset model produces three bounding boxes despite the presence of a single object. One of them roughly aligns with the cow but consistently cuts off portions of the animal, revealing systematic false negatives due to poor localization. Another box is excessively large and includes large regions of sky, introducing substantial false positives, while the third box does not contain any part of the cow at all yet is still classified as "cow," illustrating a failure in objectness estimation. Even the "best" box in this case is only relatively better because it overlaps the cow more than the others, not because it truly achieves acceptable localization. This example shows that even in a simple scene with a single, unambiguous object, the trained model produces noisy and unreliable detections, whereas the overfitted model reproduces a perfect box because it memorized the sample.

The second image, containing several people and a car with variations in orientation, appearance, and partial occlusion, makes the behavior even clearer. The overfitted model again produces near-perfect detections on the objects it has memorized, although it still adds extra boxes on nonexistent people or cars and misses the partially

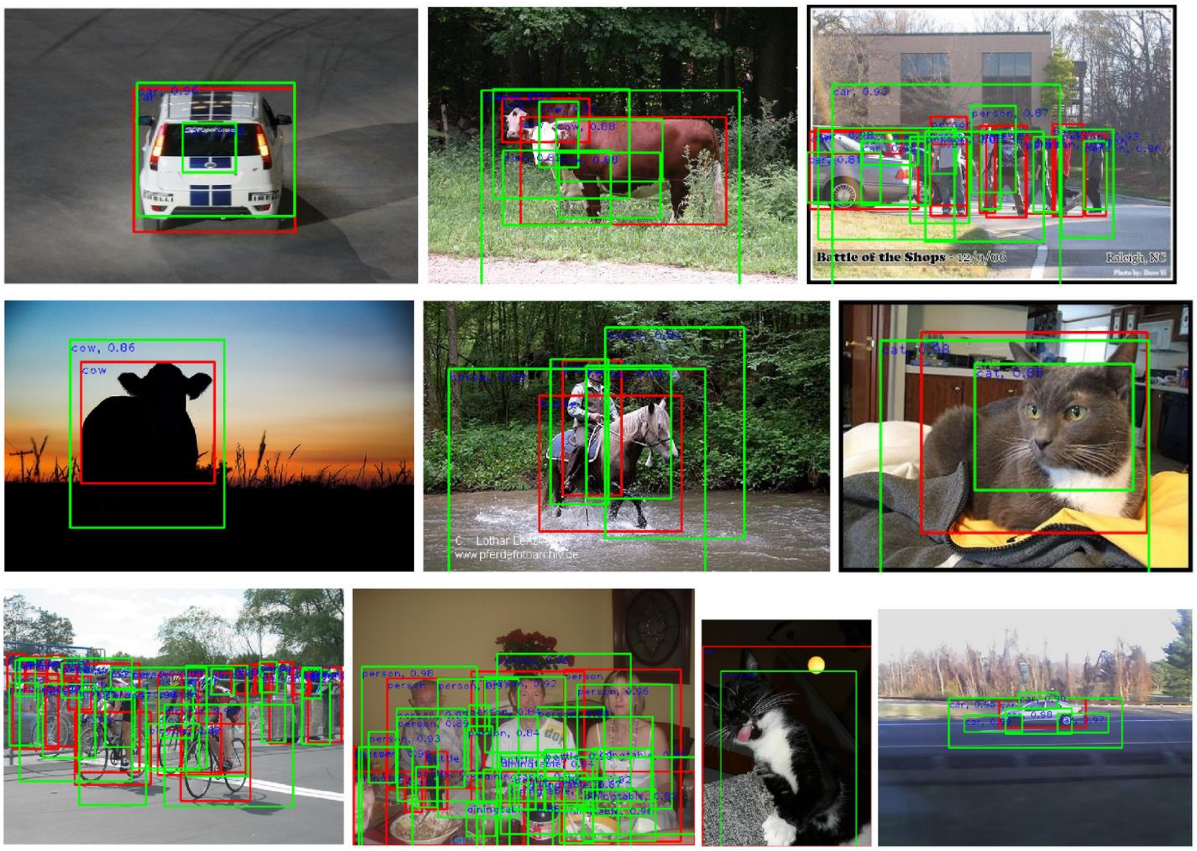
occluded vehicles on the side. In contrast, the fully trained model amplifies the failure modes seen in the cow example: it generates bounding boxes that are either too large or too small, that do not align with the true object extents, and that often overlap each other without being centered on the actual person or car. Typically, one box is somewhat closer to the true size of the person while a smaller one sits on top of it, but neither is correctly placed over the object. These observations illustrate that the primary limitation of the general model is not class prediction—which is often correct—but rather poor objectness scoring and spatial localization, resulting in false positives, false negatives, and low-IoU proposals even in relatively clean scenarios.

- For the test using the implemented Focal Loss and 50 epochs
    - Attach the detection examples for the overfitted model



The results oft the overfit model depend on what images it has seen to learn it's classes. For most objects, predictions are terrible because the model has not seen any or only a few samples describing the object (e.g. person or horse). Vehicles such as cars or bikes seem to be classified well because the model saw some samples of it during training.

    - Attach the detection examples for the model trained in the complete dataset



Obviously, the Focal Loss based model also learned to detect all the objects pretty well. Some of the boxes are too big or too small but still have sufficient IoU tob e considered as prediction. Of course, there is often some small margin that can not be reached because of interpretability reasons. One example is the shadowed cow: The GT is a pretty small box but the predicted box is bigger. The box part at the bottom of the GT could be bigger because the legs of the cow are not visible due to the shadow.
The performance of Cross Entropy (CE) Loss and Focal Loss (FL) approach are pretty similar. Some pictures contain better detections using CE, some pictures contain better detections using FL.

- Complete the following table with the mAP results proposed at the end of the notebook

	Regular YOLO loss		Focal Loss
	With positive sample filtering	Without positive sample filtering	Without positive sample filtering
mAP (50 epochs)	12.3%	11.33%	11.51%

The experiment evaluated three YOLO training configurations: Regular YOLO loss with positive sample filtering, which uses the standard YOLO loss (localization, confidence, classification) while selecting only truly relevant anchors to reduce training noise, yielding the highest mAP of 12.3%. When the filtering was removed, the Regular YOLO loss without filtering suffered a performance drop to 11.33%, confirming that training on an excess of irrelevant or ambiguous positive samples significantly degrades learning. Replacing the classification loss with Focal Loss without filtering resulted in a slight performance increase to 11.51% compared to the unfiltered YOLO loss,(The hyperparameter of focusing_param was also tried with a value of 7 and 3 with worse mAP of 10.37 and 10.45) indicating that Focal Loss helps manage class imbalance but does not fully compensate for the lack of proper positive sample selection.